Galaba Vamsi

Mechatronics at IIT Bhilai. Full-stack, AI/ML, and open-source engineering.
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SUMMARY

Graduating May 2027. Co-founder of Human Slop (Expo/React Native app, Fastify API, PostgreSQL; live on the web and Google Play). I build full-stack apps, AI agents, and ML models (Charter, Pehchaan, Kaggle and Amazon ML entries). 15 merged pull requests to open-source robotics tools; I wrote and maintain inspect-robots-wandb on PyPI.

EDUCATION

B.Tech. in Mechatronics, Indian Institute of Technology Bhilai | **Coursework:** DSA, OOP, AI and ML, Deep Learning for Computer Vision, Digital Image Processing, Optimal Control, Advanced Control Theory. Aug 2023 – May 2027

EXPERIENCE

Founder, Robotropic Physical-AI startup: safe, recorded robot-skill changes | Website Oct 2026 – Present

Co-founder, Human Slop Anti-AI social app | Web app | Play Store 2026 – Present

- Co-building it with my co-founder: an Expo (React Native) app for web and Android, a Fastify API, and PostgreSQL on Supabase, deployed on Render and Cloudflare. Live on the web and Google Play (50+ downloads).
- Added a device biometric check at signup and a keystroke-timing score in the editor; both are heuristics, not proof of authorship.

Research Intern IIT Bhilai | Advisor: Dr. Arzad Alam Kherani | RIS-SIM v2 | v1 May 2025 – Jul 2025

- Built RIS-SIM, an open-source Python emulator for RIS-assisted radio links with USRP-style SDR interfaces; designed RIS unit cells in CST Studio Suite (IEEE INDICON 2025, co-author). Rebuilt it in 2026 as RIS-SIM v2 (6 scenarios, 20 Hz WebSocket dashboard, 103 tests), then as sremu (Projects).

OPEN SOURCE

HFlow Hebbian Robotics (YC S26) | Repository | My PRs Sep 2026 – Oct 2026

- 8 merged PRs to an SDK that verifies robot training data: conversion now keeps data whole or refuses it with a clear reason, e.g. depth videos that would silently drop from 12-bit to 10-bit (#400). A benchmark PR (#367) reported no safe speedup and changed nothing.

Inspect Robots and inspect-robots-wandb Robocurve (YC S26) | Repository | PyPI | Plugin source Sep 2026 – Oct 2026

- 2 merged PRs: a configurable LLM retry policy across 5 wire protocols (#476) and unknown git state no longer logged as clean (#475). 3 open, including checkpoint/resume (#514).
- Wrote and maintain inspect-robots-wandb, a community W&B plugin on PyPI: 100% coverage gate, CI on Python 3.10–3.13, trusted publishing.

Cerulion Open-source Rust robot runtime | Repository Oct 2026

- 5 merged PRs: 1 CLI fix (#216) and 4 documentation/comment corrections.

PROJECTS

Charter (2026) Razorpay AI Buildathon prototype: chat, voice, and MCP (10 tools) share one bounded commerce core with frozen INR quotes, Razorpay test-mode payments, audit trails, and row-level security (TypeScript, Fastify, PostgreSQL). Live (test mode) · GitHub · Video

Pehchaan (team of 2; top 10 of 4,723, RAKSHAM (IIT Delhi × Amazon), 2026) UPI voice-clone fraud defence; I led ML and backend. Fine-tuned an XLS-R + AASIST detector: 4.8% EER vs 34.3% off the shelf. A real phone-call test exposed a gap (24.6% EER); modelling the speakerphone channel and retraining cut it to 6.8%, raised clones caught from 48% to 75%, and flagged 0% of genuine callers. In simulation the risk engine holds 73% of scams vs 0.21% of genuine payments (p95 3.8 ms).

sremu – Timed RIS Control Emulator (2026, MIT) IQ-level RIS emulator: timestamped commands act on an exact, logged sample; deterministic and real-time controller-in-the-loop modes; runs unmodified GNU Radio 3.10; ~2,100 tests; benchmarked on AWS. GitHub

Arista Wi-Fi RRM (team; Inter IIT Tech Meet 14.0, 5th place, Dec 2025) Client-aware radio resource management in NS-3 with I/Q-based interference classification (STFT, DBSCAN, Dual-CNN; 16 classes) on a 5-AP, 50-client topology. Slides

Open-PyFX (2026) GPU video-effects app in Python, ModernGL, and GLSL: CRT emulation, ASCII edge detection, and 4K export. GitHub

Serenity (team, 2024) Flutter wellness app with an on-device TFLite mood classifier from the camera, a stress questionnaire, and exercises. GitHub

PUBLICATIONS

- **Co-author** — *Inspect Robots: Evaluating the Capabilities and Safety of Embodied AI*, arXiv:2610.06306, 2026; accepted to the SPAIS Workshop at CoRL 2026 (poster). arXiv
- **Co-author** — *Experience with RF Energy Harvesting-Driven Self-Powered RIS*, IEEE INDICON 2025.
- **Co-author** — *A systematic literature review on simulation models and deployments for reconfigurable intelligent surfaces*, Ad Hoc Networks (Elsevier), 2026.
- **Sole author (preprint draft, arXiv submission pending)** — *Timed RIS Control in the Loop: A Sample-Accurate IQ-Level Emulator with Characterized Timing*, 2026. Draft

SKILLS

Languages: Python, TypeScript/JavaScript, SQL, C/C++, Rust (open-source contributions), Dart, GLSL
Software, cloud, and tooling: FastAPI, Node.js (Fastify), React, React Native (Expo), Flutter, Next.js, PostgreSQL, Supabase, WebSockets, MCP, Docker, AWS (CDK, Lambda, Step Functions, Bedrock), Git, Linux, GitHub Actions, pytest, Ruff, mypy, PyPI trusted publishing
ML and data: PyTorch, LightGBM, OpenCV, NumPy, CuPy, TFLite, Apache Arrow, Weights & Biases

ACHIEVEMENTS

5th of 232 teams, WEAR Dataset Challenge @HASCA 2026 (Kaggle, final private leaderboard, macro-F1 0.93137) | **AIR 200** (2026) and **AIR 700** (2025; 20,000+ participants), Amazon ML Challenge (team) | **5th place**, Arista Networks Wi-Fi challenge, Inter IIT Tech Meet 14.0 (team, Dec 2025) | **1st place**, IIT Bhilai Web Development Competition | **Top 1%**, JEE Advanced | **PMSS** scholarship (2024) | **Subroto Mukherjee** Scholarship (IAFBA, 2023) | **YC Startup School**, Bengaluru (attended)